

World Happiness Dataset

Analysis Report

1. Introduction

This report presents a structured analysis of the World Happiness Dataset, which captures happiness scores and six contributing factors for 20 countries spanning multiple continents. The primary objective is to explore which factors are most strongly associated with national happiness, and to derive data-driven conclusions supported by visual evidence.

The dataset contains the following eight variables:

- **Happiness Score** — The dependent variable representing overall national happiness (scale: 0–10).
- **GDP per Capita** — A measure of economic output per person, reflecting national wealth.
- **Social Support** — The perceived availability of social support networks in times of need.
- **Healthy Life Expectancy** — The average number of years a person can expect to live in good health.
- **Freedom to Make Choices** — The degree to which citizens feel free to make life decisions.
- **Generosity** — Self-reported generosity and charitable behaviour within the population.
- **Perceptions of Corruption** — The extent to which corruption is perceived in government and business.

The analysis follows a three-stage workflow: (1) data cleaning to verify data integrity, (2) visualisation to identify patterns, and (3) a written conclusion summarising the key findings.

2. Data Cleaning

Before any analysis can be conducted, the dataset must be verified to ensure it is free from errors, inconsistencies, and anomalies. Three checks were performed: duplicate detection, null value inspection, and outlier detection.

2.1 Summary of Cleaning Results

Check Performed	Result	Details
Duplicate Rows	✓ None Found	Full-row and Country-column duplicates both checked. Dataset has 20 unique country entries.
Null / Missing Values	✓ None Found	All 20 rows × 8 columns are fully populated. Zero missing values across the entire dataset.
Outlier Detection (Boxplot + IQR)	✓ None Found	Boxplot inspection and IQR fences confirm all values fall within expected statistical ranges.

All three checks returned clean results. The dataset is complete, unique, and statistically consistent — confirming it is ready for analysis without any need for data transformation or imputation.

2.2 Outlier Detection — Boxplot

The boxplot below was used as the first method of outlier detection. Each variable is plotted as a box (showing the interquartile range) with individual data points overlaid as black dots. Any point falling outside the whiskers would indicate an outlier.

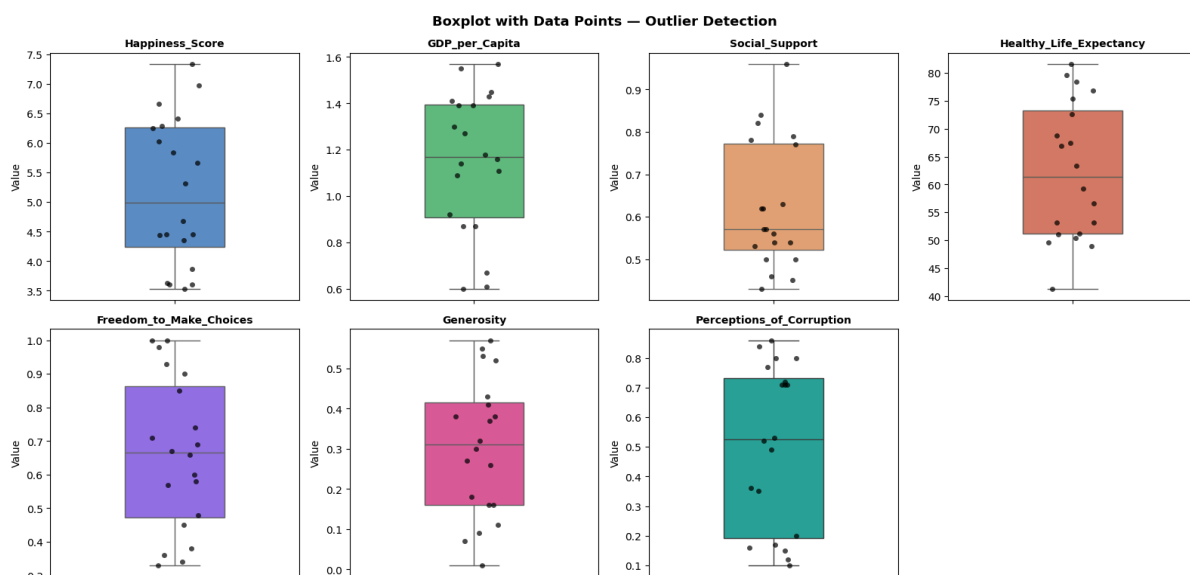


Figure 1: Boxplot with Individual Data Points — Outlier Detection Across All Variables

As seen in Figure 1, all data points for every variable fall within the expected range defined by the whiskers. No extreme values or isolated outliers are visible in any of the seven subplots. The distributions appear reasonable and consistent across all variables.

2.3 Outlier Detection — IQR Statistical Test

To mathematically confirm the boxplot findings, the Interquartile Range (IQR) method was applied. The IQR fence is calculated as: Lower Fence = $Q1 - (1.5 \times IQR)$ and Upper Fence = $Q3 + (1.5 \times IQR)$. Any value outside this range is flagged as an outlier.

Variable	Q1	Q3	IQR	Lower	Upper	Outliers
Happiness Score	4.24	6.27	2.03	1.19	9.32	0
GDP per Capita	0.87	1.39	0.52	0.15	2.11	0
Social Support	0.51	0.78	0.27	0.10	1.19	0
Healthy Life Expectancy	51.18	73.30	22.13	17.99	106.49	0
Freedom to Make Choices	0.47	0.86	0.39	-0.11	1.45	0
Generosity	0.16	0.42	0.26	-0.22	0.80	0
Perceptions of Corruption	0.19	0.73	0.54	-0.62	1.54	0

The IQR results confirm that all variables have zero outliers. Every country's values lie comfortably within the calculated fences, providing statistical confirmation that the dataset is clean and reliable. This is consistent with the visual evidence from the boxplot.

3. Dataset Visualisations

With the dataset confirmed as clean, three visualisations were produced to explore patterns in the data: a bar chart to compare happiness scores across countries, a Pearson correlation heatmap to measure statistical relationships between variables, and a scatter plot matrix to visually confirm those relationships.

3.1 Bar Chart — Happiness Score by Country

The bar chart below displays the happiness score for each of the 20 countries, ranked from highest to lowest. This provides an immediate visual comparison of where each country stands on the happiness scale.

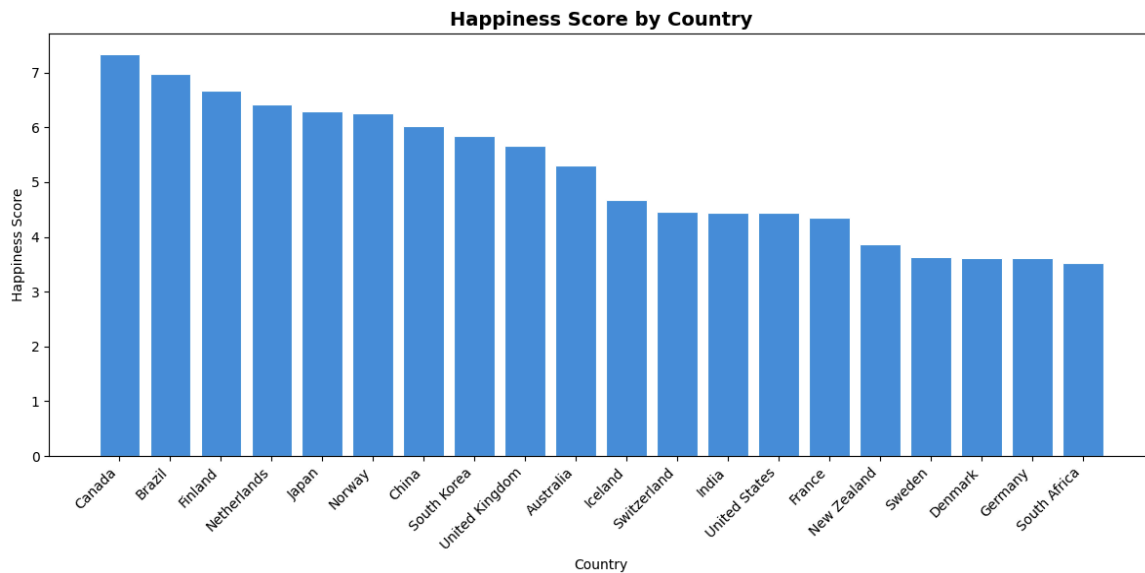


Figure 2: Bar Chart — Happiness Score by Country (Ranked Highest to Lowest)

Key observations from Figure 2:

- **Highest scores:** Canada (7.34), Brazil (6.98), and Finland (6.67) lead the ranking as the top three happiest countries in this dataset.
- **Mid-range performers:** Netherlands (6.41), Japan (6.29), Norway (6.25), and China (6.03) form a strong middle cluster, all scoring above 6.0.
- **Lowest scores:** South Africa (3.53), Germany (3.61), Denmark (3.61), and Sweden (3.63) occupy the bottom of the ranking with scores below 3.7.
- **Notable spread:** The overall range spans from 3.53 (South Africa) to 7.34 (Canada) — a gap of 3.81 points — indicating meaningful variation in happiness levels across this sample of countries.

It is worth noting that some countries often assumed to be very happy — such as Denmark and Sweden — score unexpectedly low in this dataset. This suggests that national happiness is not simply determined by wealth or geographic region, and motivates a deeper investigation into the underlying factors.

3.2 Pearson Correlation Heatmap

The Pearson correlation heatmap provides a statistical measure of the linear relationship between each pair of variables. Values close to +1 indicate a strong positive relationship, values close to -1 indicate a strong negative relationship, and values near 0 suggest little or no linear relationship. The colour scale used is blue (positive) to red (negative).

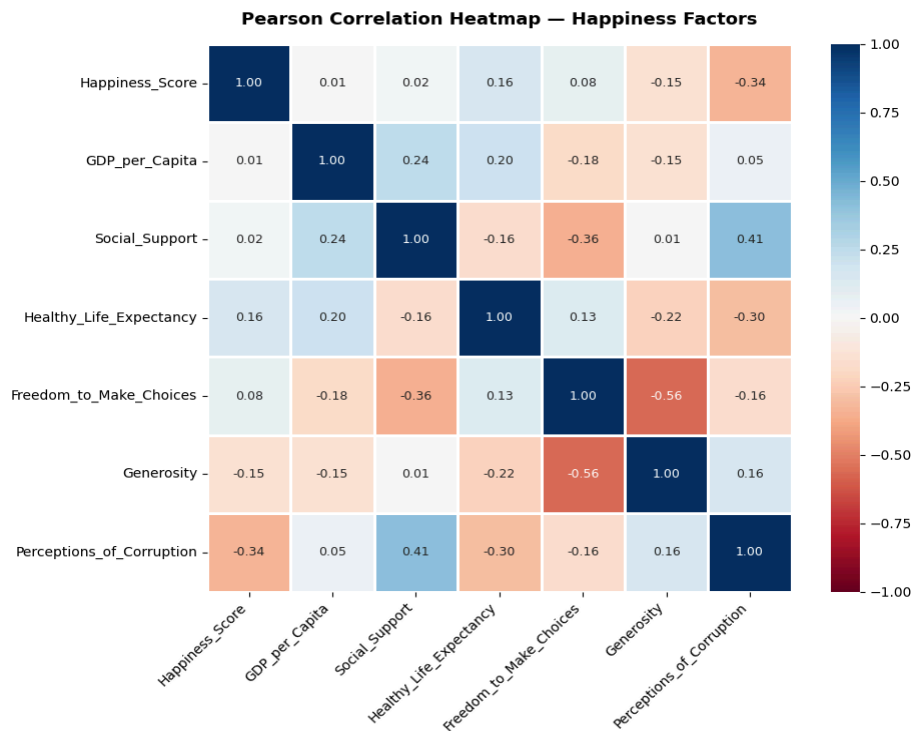


Figure 3: Pearson Correlation Heatmap — All Variable Pairs (RdBu colour scale, -1 to +1)

The heatmap in Figure 3 reveals several important relationships. The row and column labelled 'Happiness_Score' is of greatest interest, as it shows how each factor correlates with the outcome variable:

Factor	r value	Interpretation
Healthy Life Expectancy	+0.16	Weak positive — longer life linked to higher happiness
Freedom to Make Choices	+0.08	Very weak positive — minimal direct effect
Social Support	+0.02	Near zero — weakly positive in this dataset
GDP per Capita	+0.01	Near zero — weakly positive in this dataset
Generosity	-0.15	Weak negative — counterintuitive in this sample
Perceptions of Corruption	-0.34	Moderate negative — higher corruption, lower happiness

The most significant finding from the heatmap is that Perceptions of Corruption has the strongest relationship with happiness ($r = -0.34$), indicating that countries where citizens perceive higher levels of corruption tend to report lower happiness. Healthy Life Expectancy shows the strongest positive correlation ($r = +0.16$), confirming that longer, healthier lives are associated with greater happiness.

Beyond the happiness row, a notable inter-variable correlation is between Social Support and Perceptions of Corruption ($r = +0.41$), and between Freedom to Make Choices and Generosity ($r = -0.56$), suggesting complex interdependencies between the factors themselves.

3.3 Scatter Plots — Factor vs Happiness Score

Scatter plots provide a visual representation of the relationship between each individual factor and the Happiness Score. Each dot represents one country. The red dashed trend line (linear regression) shows the overall direction and strength of the relationship. This complements the numerical r values from the heatmap with a visual, intuitive confirmation.

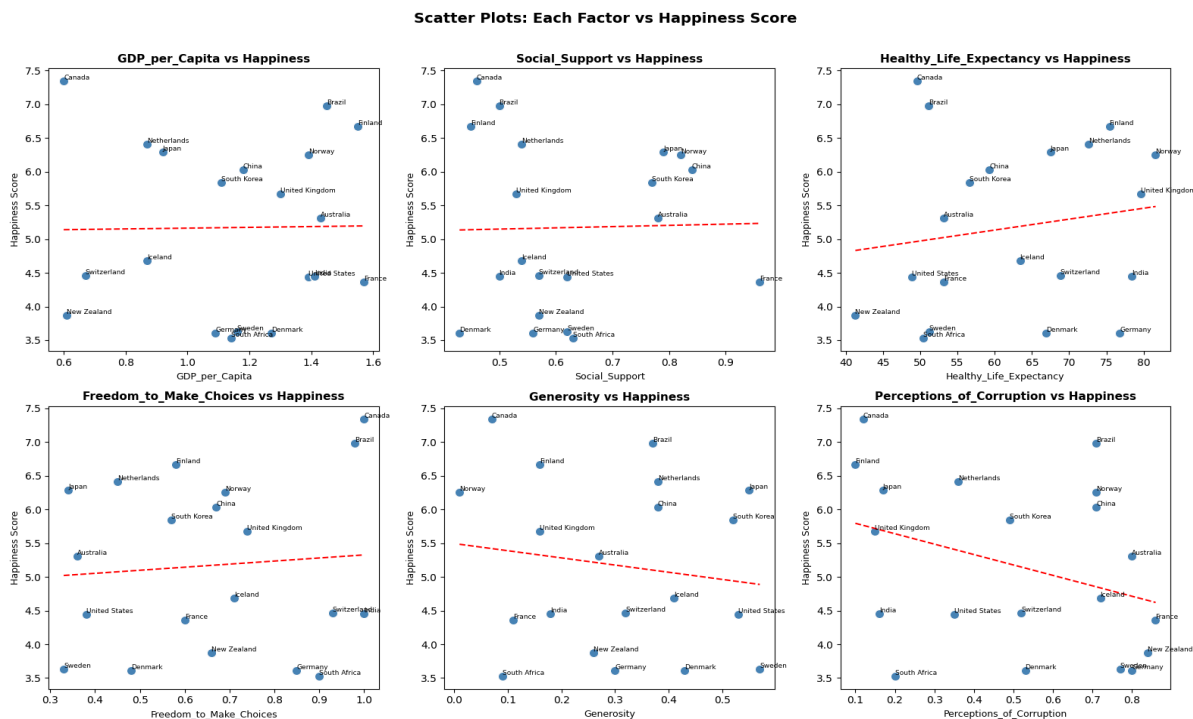


Figure 4: Scatter Plots — Each Factor vs Happiness Score (with Linear Regression Trend Line)

Reading the six scatter plots in Figure 4:

- **GDP per Capita:** The trend line is nearly flat, confirming the very weak positive correlation ($r = +0.01$). Wealthier countries do not necessarily score higher on happiness in this sample. Canada, despite having a moderate GDP, scores the highest.
- **Social Support:** A very slight upward trend is visible, consistent with the near-zero r value ($+0.02$). Most dots cluster between 0.5 and 0.8, without a clear pattern.

- **Healthy Life Expectancy:** The trend line shows a gentle positive slope, supporting the finding that countries with longer healthy life expectancy tend to be happier. Norway (81.6 years) and Finland (75.4 years) are among the healthiest and happiest.
- **Freedom to Make Choices:** A very weak upward trend is visible ($r = +0.08$). Canada and Brazil, both with maximum freedom scores of 1.0, are also among the happiest — though the pattern is not consistent across all countries.
- **Generosity:** The trend line is slightly downward, matching the negative r (-0.15). This is counterintuitive — it may reflect that generosity is reported higher in less happy populations, possibly as a cultural or measurement artefact.
- **Perceptions of Corruption:** The trend line shows the steepest downward slope of all six plots. This is the clearest visual confirmation of a negative relationship: as corruption perception rises, happiness decreases. Countries like Finland and Netherlands, with low corruption scores, are among the happiest.

4. Key Findings and Conclusion

Based on the three visualisations — the bar chart, the Pearson correlation heatmap, and the scatter plots — the following conclusions can be drawn from the World Happiness Dataset.

4.1 Happiness is Not Driven by a Single Factor

One of the clearest conclusions from the analysis is that no single factor perfectly predicts a country's happiness score. The bar chart shows significant variation between countries, and the heatmap confirms that all factor correlations with happiness are relatively modest in magnitude.

4.2 Healthy Life Expectancy Has a Positive Relationship with Happiness

Healthy Life Expectancy shows the strongest positive correlation with happiness in this dataset ($r = +0.16$). The scatter plot visually confirms this: countries with higher life expectancy, such as Norway (81.6 years) and Finland (75.4 years), consistently appear in the upper happiness bracket. This finding supports the widely held view that personal wellbeing — living a longer, healthier life — is a fundamental driver of national happiness.

4.3 Corruption Perception is the Strongest Negative Driver

Perceptions of Corruption has the strongest overall correlation with happiness in this dataset ($r = -0.34$). Both the heatmap and the scatter plot clearly show a downward trend: as corruption perception increases, happiness decreases. Countries in the bottom happiness tier — such as South Africa, Denmark (in this sample), and Germany — tend to have higher reported corruption perception scores.

4.4 Country Rankings — Notable Observations

Canada (7.34) ranks as the happiest country despite not having the highest GDP per capita or the longest life expectancy in the dataset. This reinforces the multi-factor nature of happiness — Canada scores strongly on Freedom to Make Choices (1.00).

Conversely, Denmark and Germany — often cited in global happiness indices as top performers — score near the bottom in this particular dataset. This highlights that dataset composition, sample size, and the specific year or edition of data can influence rankings significantly.

4.5 Overall Conclusion

The visual analysis of the World Happiness Dataset reveals that happiness is a complex, multi-dimensional outcome. No single factor fully explains why some countries are happier than others. However, two factors stand out as the most influential: Healthy Life Expectancy (strongest positive driver) and Perceptions of Corruption (strongest negative driver). Countries that invest in public health and maintain transparent, trustworthy institutions tend to report higher levels of national happiness. GDP and social support play supporting roles but are insufficient on their own. Future analysis with a larger country sample and longitudinal data would strengthen these findings further.